

Risk assessment methodology for network integrity

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Modern, high functionality networks support an ever-increasing range of sophisticated services; but this strength is also a weakness due to the fragility of the complex distributed-processing capability. Unexpected perturbations can cause widespread network outages, i.e. 'brownout'. The risks to network integrity are exacerbated by regulatory policies that demand open network interconnection between competing network operators and service providers; and the increasing complexity where single experts cannot comprehend the whole problem space. This paper proposes a design methodology for a formal and systematic framework to assess the risk to network integrity and hence minimise network and service failure probability.

1. Introduction

Telecommunications networks are becoming increasingly complex. The introduction of processor controlled digital systems with message-addressed common-channel signalling [1] permits the introduction of the intelligent network (IN) [2, 3] to satisfy customer demand for rapid deployment of increasingly sophisticated services. However, such a fusion of telecommunications and computing into a distributed processing environment gives rise to a dramatic increase in control complexity and fragility. Services have to co-exist and co-operate on different platforms sometimes giving rise to undesirable feature interactions [4–9]. Unexpected perturbations may also ripple through the network to give widespread outages, i.e. the so called 'brownout'.

These problems are exacerbated by regulatory policies that demand the open interconnection of networks of competing operators and service providers together with internetworking of services, e.g. the ONP (Open Network Provision) policy of the European Union [10]. The threat to network integrity is high in such an environment and its preservation is of paramount importance. The traditional manual approach, however, to the maintenance of network integrity which relies on the knowledge of individual experts is no longer valid in such a complex environment. There is a need to develop a formal and systematic methodology to assess the risk to network integrity — a framework to pre-empt, detect and quickly resolve failures in telecommunications networks and the services they support. The importance of finding mechanisms to protect the integrity of the networks is recognised by many operators. In particular BT, as part of the investigation in

this area, is funding a research project associated with the work presented in this paper.

2. Network integrity definition

Network integrity has been defined as: "...the ability of a network to retain its specified attributes in terms of performance and functionality"[11]. Network integrity is then the ability of a network to maintain a safe state in which it is invulnerable to unexpected perturbations. Therefore it can be said that a network possesses a high degree of integrity when it is secure and robust against adverse conditions, e.g. signalling overload. If the network is in a state of high integrity, these perturbations can be easily 'absorbed' by the system without having undue effect. Alternatively, if the network is in a state of low integrity, the same perturbations might have more serious consequences, even resulting in failure.

Figure 1 shows that the degree of integrity can be divided in a collection of states of integrity. At one end there is 100% integrity, which means that the system is absolutely robust and will not be affected by unexpected problems. At the opposite end, there is 0% integrity, meaning that any perturbation will cause failure. Between those two limits, the network can be at different states of integrity.

The different bands between 0% and 100% of integrity should be represented as a continuous gradient. However, this is not feasible in practice. Due to the large complexity

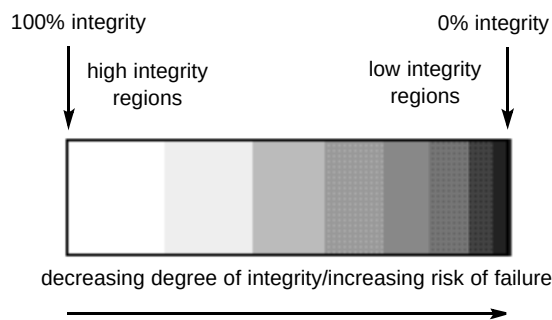


Fig 1 Integrity definition example.

of the problem, the parameters introduced to measure integrity must take discrete values, and hence the need to introduce real boundaries between bands with different values of integrity. The separation between these bands will not always be strict and there may be overlaps.

Breach of network integrity and failure are related concepts, but network integrity goes beyond failure identification. Loss of integrity means moving from one band to another band of lower integrity, but this does not necessarily imply failure. Breach of integrity would imply failure only in one of the following cases:

- if the system was initially in a band of very low integrity, because any minor perturbation would put it in a situation of failure — hence, it is important to operate in a high-integrity band,
- if the loss of integrity is very high, causing the system to move from a band of acceptable integrity to the region near 0% — in some cases loss of integrity cannot be avoided, e.g. there may always be unforeseen problems which, depending on the state of the system might have a chain reaction effect, leading to serious failure, but it is desirable to minimise such instances by identifying possible sources of failure and designing them out, or monitoring the relevant parameters to take avoiding action.

It is therefore necessary to develop a framework to measure and categorise the degree of integrity of a network into the spectrum of bands. Such a framework must:

- discriminate between different states of integrity, or bands of risk, and permit the identification of which state the network is operating in,
- allow the assessment of the probability of moving from one band of risk to another,
- be formal and well defined to aid understanding.

The construction of such an integrity framework includes:

- specifying a definition of network integrity, examining potential causes of loss of network integrity and how this loss would manifest, i.e. which parameters must be considered as indicators of the degree of integrity of a network,
- classification of the degree of integrity in regions or bands, distinguishing between safe bands, bands of considerable danger, very dangerous bands, etc — the borders between these bands cannot be strictly defined, but it should be possible to identify ranges within which the different parameters can vary.

The actual shape and size of the integrity bands will be determined by complex relationships between the different integrity parameters, and will be different for different systems. The example in Fig 1 represents only a very simple case, in which it is assumed that the size of the bands decreases towards the lower integrity regions. This represents the fact that the risk of failure increases as the degree of integrity decreases. Making the integrity bands narrower in the low integrity regions allows the detection of small integrity fluctuations. Conversely, in the high integrity regions, small changes in the degree of integrity will not have distinct effects, and therefore the integrity bands can be wider. This approach encourages a rational prioritisation in the allocation of resources and investment in the maintenance of the network integrity. If a network is operating in a high integrity band, the effort and cost of detecting minor integrity deviations might be much bigger than their effects. On the other hand, the costs associated with the restoration of integrity, e.g. human resources, equipment, inaccessibility of services during repairing time, etc, will be larger if the network is operating with a very low degree of integrity. Hence the need to avoid large breach of network integrity in the low integrity regions.

The example described above is a simplified case based on a one-dimension definition of network integrity. A more realistic view is a multidimensional definition, in which each dimension is given by a category of integrity parameters, with their own integrity ranges.

The definition of the integrity bands will be different for each particular system, i.e. each operator will define their own criteria. What is necessary is a general framework that is commonly understood and can be customised for different scenarios. These criteria can be variable, hence the boundaries between bands of integrity will not be static; it must be possible to redefine them according to the current criteria.

In order to develop an integrity framework such as the one described above, the following actions are necessary:

- identification of the parameters that define the degree of network integrity,

- identification, for those parameters, of the threshold values that would indicate a significant loss of network integrity,
- finding an efficient way to monitor and process that information.

Network integrity is a relatively new topic and there is limited literature relating to the subject. However, it has parallels with many different areas, e.g. signalling, testing, management, etc, where there is expertise, but it is isolated and does not integrate easily into a coherent approach.

A necessary initial stage is a compilation of information. Effort must be made to integrate these isolated branches of expertise under the umbrella requirements of integrity. Such an expertise base will be well beyond the comprehension of one individual. Therefore it is of paramount importance that it is incorporated into a meaningful and manageable structure, easily accessible by a variety of specialists.

3. Requirements of a network integrity measure

This section describes some of the requirements for a framework to define and measure network integrity in telecommunications networks.

3.1 Flexible

The methodology must embrace a wide diversity of the telecommunications scenario, where an extensive variety of elements have to interwork, e.g. interconnected networks of different operators, equipment and services from different providers, and multiple interacting and internetworked services. The framework must be flexible, allowing differentiation between the common areas and specific situations.

3.2 Formalised

The framework must be formal and well defined so that it can be commonly understood, and numerical or semi-quantitative, to permit the introduction of comparative measurements. It must allow for discrimination between different bands of risk and place a network in one of these regions, and assess the probability of moving from one band to another. Rather than providing absolute values, it must permit the identification of an increase in risk, based on comparison between the measured values and the defined thresholds. Thus it is necessary to define the thresholds that separate different regions and identify the critical values that imply high risk.

3.3 Structured

Because of the complexity and diversity of networks, the threats to integrity are large and varied. Hence there is a need to decompose problems. This requires a hierarchical framework where the top level contains parameter groups, describing different aspects of the problem, which can be progressively decomposed into more detail in the lower levels. Relevant paths through the hierarchy can be followed for particular applications.

There is also a need to decompose the integrity of a whole system into different levels of integrity, i.e. unit level, integration level, system level, service level, etc. Also, for each level, the integrity of various components needs to be examined, e.g. integrity of the data, integrity of signalling messages, etc.

3.4 Adaptable and expandable

To cater for the continuous and rapid change in telecommunications, the network integrity framework must be adaptable and expandable to permit the incorporation of new features. Adaptability and expandability are also required to provide for a dynamic and progressive model development, and to allow upgrades to account for new parameters and relationships identified by 'trial and error'.

3.5 Focused on services

The evolving telecommunications market demands the rapid deployment of advanced and increasingly complex, IN-based, services that need to co-operate across different platforms. Loss of integrity may affect the various services differently and discrimination is required to enable risk to individual services to be separately evaluated. This requires an understanding of how services/features operate and interact with network entities and other services/features.

The following needs consideration:

- internal behaviour of services, e.g. databases accessed, types of signalling transactions, concentrations of signalling activity, etc, to identify the critical points in the network for particular services,
- interactions between:
 - different services/features,
 - services and network entities,
 - different network entities,
- usage characteristics, interactions of the users with the services,

- performance of the systems, that can impose constraints and limitations to the service behaviour and even cause failure.

4. Integrity factors

The following are some of the factors or areas that should be considered in order to tackle the network integrity problem.

4.1 Population and user models

The behaviour and operation of a service is strongly affected by the way the service is used by the customers. These factors can be embraced in so-called 'customer models'. These customer models can be decomposed in two main areas, that are called user models and population models. User models represent how an individual user would interact with the service, e.g. what actions they are likely to take, possible errors they would make and how they would react in different situations. Population models represent the characteristics of the whole collection of users of the service. This includes, for example, number of customers, their geographical distribution, their degree of mobility, and their usage patterns, e.g. call-holding times, average and peak calling rates, etc. All these factors need to be carefully considered when a service is designed and implemented. A service may be conceived to have two or more implementations, e.g. a demonstration version, used when the service is being introduced to capture market share, and a more realistic version to be used as service popularity arises.

4.2 Quality of service

Quality of service (QoS) represents how the service is perceived by the customers. In a market environment such as telecommunications, where customer satisfaction is a primary objective, special care must be taken to understand what parameters specify customers' perceptions, and to maintain the appropriate values for them. A major problem is how to map the users' QoS requirements into network parameters, and a great deal of work is being done in this area, e.g. the RACE project TOMQAT [12], and the ACTS project MISA [13].

4.3 Performance

The behaviour of a service can be heavily influenced by network performance characteristics. Therefore, the study of possible erroneous behaviour must include the relationship to performance.

4.4 Signalling

This factor refers to the general information flows within the system, as opposed to specific signalling protocols or architectures. Signalling is the area of most growth in complexity, and hence highest risk to integrity, in the new telecommunications scenarios. For this reason, it is a factor that must always be carefully analysed.

5. Required Activities

In order to tackle the problem of maintaining integrity of new advanced intelligent communications systems, emphasis must be given to three main activities, namely, modelling, risk analysis, and documentation techniques.

5.1 Modelling

Modelling and simulation techniques are needed to improve the understanding of the new systems before they are developed for real. This is particularly important in the telecommunications environment, which is characterised by fast changes and increasing complexity.

5.2 Risk analysis

A great part of the problem of maintaining network integrity is being able to identify areas of risk, and predict the consequences of actions such as limitations and constraints of the developed systems. Consequently, risk assessment methodologies should play a main role in the integrity problem.

5.3 Storing and using information

As systems evolve and become more complex, the amounts of information produced during their development and subsequently required to understand them increases, and the traditional manual approach to the maintenance of network integrity, which relies on the knowledge of individual experts, is no longer valid. Hence the need to store this information in a structured and automated way that makes it accessible, meaningful and useful to different people, becomes apparent.

Knowledge-based systems (KBS) containing records of previous experiences must be developed. The technical requirements of such a system, such as time response, capacity, data structure, knowledge paradigm, etc, must also be carefully analysed.

6. Methodology framework

A methodology to control the degree of integrity of a network requires the use of various disciplines and must be applied coherently through the whole development cycles of the services operating in the network. The methodology framework has been divided into two categories, namely static and dynamic actions. Static actions are previous to the launching of the services, i.e. actions taken during the design and implementation stage. Dynamic actions are those actions taken in real time, i.e. during the operation of the services. Section 2 stated the need for a structured definition of network integrity, identifying the integrity parameters and their thresholds, and allowing to categorise the degree of integrity of a network. The areas of use of such a definition and the disciplines that need to be incorporated are described next.

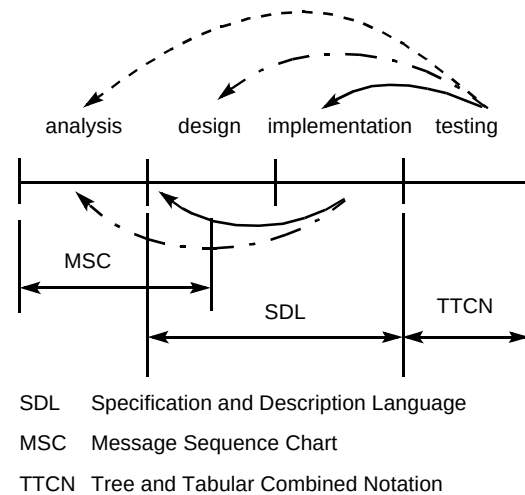


Fig 2 The design cycle and languages commonly used at each stage.

6.1 Static actions

Improvement of design methodologies

Systems that are robust to loss of integrity require design methodologies and principles that take account of factors affecting integrity.

Figure 2 shows the different stages involved in the development of a system, namely analysis, design, implementation and testing, together with some of the formal techniques commonly used in telecommunications systems at each stage, namely SDL, MSC and TTCN [14–16]. The arrows in the diagram represent the feedback loops between the different phases. Ideally, a thorough and complete analysis followed by a correct design would lead to the implementation of a system that works as expected. In reality, it is virtually impossible to implement a system that works perfectly at the first attempt; hence the importance of testing [17]. In Fig 2 testing has been included as a separate activity at the end of the development cycle for convenience, but in fact, it spans along all the steps of the development cycle.

Errors may appear at any stage, and their detection may make it necessary to go back to previous phases and introduce modifications. Feedback between the separate phases is inevitable, but a good methodology should help to reduce the long jumps, e.g. to avoid modifying the analysis or the design after the implementation phase has been completed. It is very important for the development of an efficient design methodology to have available information concerning where and how things can go wrong. In order to do this, information needs to be extracted from the cycle described above and documented in a way that can be used for other applications.

Static risk analysis

Risk analysis must be integrated into the development process, in order to understand the limitations imposed in the specification, design, implementation and testing of systems and the risk associated to these limitations. One of the reasons for these existing risks is the trade-off between the robustness of a system and the associated cost. Building a system that is absolutely robust and designed to work under any circumstances is not feasible for two reasons. Firstly, the high cost associated with it, and, secondly, it is not possible to predict all the conditions under which the system will have to work in the future. This is particularly true in the rapidly changing telecommunications environment. For this reason, it is necessary and inevitable to impose constraints and limitations to the systems — necessary, in order to reduce costs, inevitable because, even if the systems are designed to work for a wide spectrum of situations, it cannot be assumed that unexpected situations, with which the system cannot cope, will not happen in the future.

These types of limitation also exist in the testing arena. Due to the high complexity of the services, it is not possible to test everything. Testing must focus upon those areas where problems are more likely to arise, whereas other aspects that can be assumed to be non-dangerous will not be included in the testing. The level of testing carried out must also be subjected to risk assessment.

Since it is clear that these limitations will exist, it is of paramount importance to understand what their consequences are. For this reason, a risk assessment methodology that helps to assess the probability, impact and consequences of actions must be developed and integrated in the development of systems.

Part of the risk assessment activity would consist in keeping records of any constraints decisions taken during the development of the systems, and an assessment of the effects that these limitations would have in adverse conditions. Such an activity would:

- provide a better understanding of the systems operations,
- make the information accessible to different people, instead of being only in the heads of those involved in the design process,
- help to identify weak points in the systems and where problems are likely to arise,
- help in the diagnosis of failure and to speed up the identification of the actions required to fix the problems,
- help to evaluate consequences of failure, for example, in terms of lost calls, damage to equipment, cost, etc.

6.2 Dynamic actions

Monitoring and control

Monitoring and control of the integrity parameters is required to identify the band of integrity in which the network is operating at any time, i.e. if it is in a safe state or, conversely, in a situation of high risk, where unexpected actions or modifications, such as the introduction of a new service or network interconnect, incurs a high probability of catastrophic failure. The framework can be used to predict the loss of integrity due to such actions and provide a criteria to take appropriate measures, e.g. not to allow the interconnect in the terms proposed.

Restoration mechanisms

Due to the high complexity of telecommunications networks, the possibility of failure cannot be completely eliminated, even with a rigorous design methodology and exhaustive testing. Therefore, the design of efficient recovery mechanisms in case of integrity violation is as important as the prevention of failure. For this purpose it is necessary to have a good knowledge of the system behaviour together with information that helps to diagnose the cause of failure. Monitoring the integrity parameters in real time will permit rapid detection of threshold violation. Information about possible error states and their probabilities can be used to rapidly identify the source of failure and trigger the pertinent actions. These actions will depend on the criteria set by 'operators', depending on the band of integrity in which the system is operating.

Dynamic risk analysis

Risk analysis is also necessary during the operation of the services, in order to assess the severity of detected integrity violations and to decide the appropriate restoration policy to be taken. For high integrity violations, the recovery actions might involve a complete reset of the system, whereas minor integrity deviations might be acceptable if the cost associated with restoring integrity exceeds the cost due to the integrity loss. These decisions must be taken based on a rigorous understanding of the risks taken; hence the need for dynamic risk analysis.

7. Modelling methodology

New technologies and approaches in telecommunications, such as intelligent networks, will permit the fast provision of services and a large increase in the number of different services offered. These facts make it more and more difficult to comprehend and understand the behaviour and operation of services by single individuals. As a result, more structured, formalised and automated approaches are required.

In this complex scenario, modelling becomes an activity of paramount importance. The reasons for this are:

- reduction in costs, i.e. the cost of the resources required for modelling is much lower than those of the real equipment that the launch of the service requires,
- an adequate modelling exercise helps to detect and resolve problems before the launching of the services,
- telecommunications markets are moving towards new services where there is no expertise; appropriate modelling can provide a better understanding of these new services — it is worth remarking that the value of a model is not only the results it produces, but the expertise and knowledge acquired during the construction of the models.

As quoted in Woollard [18], in order to build a good model, the issues of abstraction, completeness and simplicity of the models need to be carefully considered. The models must capture all the aspects of interest of the problem they are trying to solve, but at the same time maintain a level of simplicity that allows the models to be understood and the results interpreted. This trade-off between the simplicity and the accuracy of the models requires that the areas of relevance to the problem are identified, and those aspects that would not bring any further information are disregarded.

Due to the large and increasing complexity of services, it is necessary to apply some decomposition paradigm that allows the problem to be broken down into smaller areas. A top-down approach is proposed [19], that borrows heavily

from the IN conceptual models, consisting of four main layers corresponding to different levels of detail — service layer, functional layer, operational layer and physical layer. Additionally, there is a customer layer that represents the way the users interact with the service (see Fig 3). The important point of this decomposition is that different aspects of the models would access only those layers that are required, thus providing a framework that allows different aspects of the problem to be treated with the level of detail needed:

- **service layer** — comprising the set of features representing the service, which can be modelled as a set of interacting objects,
- **functional layer** — comprising the functionality that supports the service, how the functions are provided in the network and their interaction to provide the required behaviour,
- **network layer** — comprising the network resources used by the functional entities, and the distribution of these resources in the physical network,
- **operational layer** — in which each entity is broken down into its operating components,
- **customer layer** — representing the interaction of the users with the service, i.e. user and population models.

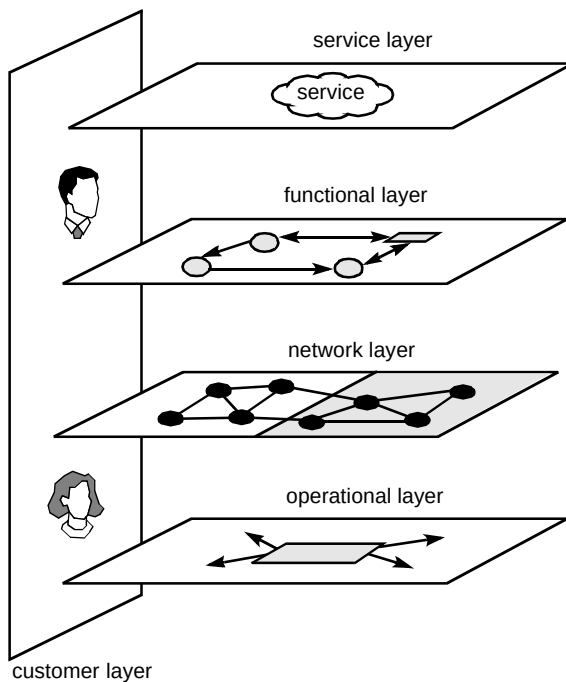


Fig 3 Layered modelling framework.

A specification language is needed, and SDL (the ITU Specification and Description Language) [14] is already in widespread use. The advantages of SDL are that it is an international standard, it is easy to use, it has proven to be the most commonly used method in telecommunications

applications (currently used by a large number of operators and manufacturers, including BT), and there are advanced software tools available to build and test SDL systems. Moreover, there is a strong international research activity in SDL, and modifications are introduced to the language to respond to new demands. Hence SDL is a likely candidate to be used as the modelling language.

8. Example service

A preliminary modelling exercise has been carried out for several purposes, namely:

- to put into practice the modelling methodology and test its suitability,
- to use it as a first approach to understand service operation and identify service features,
- to analyse which of the identified service features are relevant in the context of integrity,
- to begin to understand the influence of population models in the response and way of operation of the service.

8.1 Brief description of the model service

The objective of the modelling work is a generic representation of management of location of resources within a network. The model service customers have full mobility capability, and different mobility patterns were considered. The location of resources available for each user is described in a user profile, which is always moved to the user's local node. The resources in this case are data files whose location is not fixed. Several file relocation strategies were considered in the exercise.

At the service level, the model service can be viewed as composed of three main features:

- an addressing function, that deals with the searching of the requested data files within the network,
- database consultation, or file access function, to access the requested information in the data files,
- a file transfer function, that performs the relocation of the files.

The application of the proposed modelling methodology, for the particular scenario of two interconnected networks providing the model service, leads to the decomposition represented in Table 1, taking into account that the object of the study is the network integrity due to interconnect.

Table 1 Decomposition of the example service.

	Addressing function	File access function	File transfer function
Functional layer	Might cross-interconnect. Requires more detail.	Local function, but response time may be critical.	Treated as reliable. May need to include failure rates.
Network layer	Need to identify addressing entities and their interactions.		
Operational layer	Type of addressing scheme.	Type of file access system, response times, etc.	
Customer layer	Number and mobility of users determine addressing scheme.		Mobility of users and size of files determine relocation strategy.

8.2 Problems due to interconnect

The regulatory policies that attempt to encourage competition, e.g. the European Union Open Network Provision [10], will demand different types of interconnection, between networks and services. The interconnect point is going to be a point of weakness and a possible source of problems that can jeopardise the integrity of the interconnected networks/services. A main cause of this can be incompatibilities in the software used at either side of the interconnect and wrong translations in the boundary. For example, if two networks use different signalling systems, or even different interpretations of the same standard, this can give rise to misinterpretation of signalling messages which might have disastrous consequences.

In the proposed case of two networks providing the model service, several integrity issues arise. Taking the addressing function as an example, at the functional level, it can be identified that the addressing function might cross the interconnect, if the resource requested from one network is present at the other network. Therefore, a higher degree of detail is required. At the network layer, the entities in the system that provide the addressing function and how they interact must be identified. In the example service there is a central database containing the list of resources and their current locations. At this point, the customer layer needs to be accessed to understand possible limitations to the service. For example, if the number of customers is very large, or they have a high degree of mobility, a centralised structure is not appropriate, because a very large number of updates to this data structure would be required. In this case, a hierarchical addressing scheme would be more appropriate. Conversely, if the users have a very low degree of mobility, searching procedures based on flood filling could be used, because the number of searching messages would not be too large. If the two interconnected networks

have different addressing functions, integrity problems may arise. Hence the need to model the addressing function at the operational layer, to understand how the addressing scheme is implemented.

8.3 Key issues identified

The behaviour of the service was studied in terms of performance and QoS. The results are outside the scope of this paper, but the main value of the exercise was the understanding and expertise gained from it. In particular, it served to reinforce the following ideas:

- the importance of signalling in the integrity,
- the relevance of population patterns,
- the importance of performance characteristics.

9. Proposed approach

9.1 Overview

The initial approach focuses on ‘signalling’ (defined as the general information flows within the system and the associated processing). The approach is from a services perspective, where a service is viewed as a collection of distributed entities performing different tasks and interacting with each other to provide the required functionality. The aim is to identify the possible states of the system, the transitions between them and their probabilities, both for normal behaviour and possible error states.

The behaviour of a system can be represented as an extended finite state machine, as is done in SDL [14]. A system run or trace is a specific path of behaviour, i.e. a set of states and the transitions between them. A path of correct behaviour consists of a sequence of acceptable states. Under adverse conditions, the system might leave this path and move to an erroneous state. Some errors will not produce a strong deviation from the path of correct behaviour and therefore can be considered minor errors, since the system remains within an area of acceptable behaviour. Conversely, major errors or a combination of errors can cause ‘control mutations’, i.e. can make the network follow a path divergent from that of the correct behaviour. The aim of the framework is to obtain the overall probabilities for complete paths of behaviour, i.e. the probability of moving from one initial state to any of the possible final states. It must provide the probability of erroneous behaviour and should identify the risk of control mutations which can take the system out of its stable condition towards catastrophic breakdown. There is a need to detect deviations and their size, classifying regions of risk and categorising errors according to their consequences. Thus minor errors can be

treated differently to those leading to a catastrophic situation.

The concepts are illustrated in Fig 4, where the large triangle represents the whole domain, i.e. all the possible paths that the system can follow when the service is running, the shaded area indicating the region of acceptable behaviour. The path of correct behaviour, starting from the initial state (the top vertex), is shown, together with minor deviations that return to the correct path (Type 1), or are detected and cause the process to stop (Type 2), or terminate in a final state out of the normal path but still within the safe region (Type 3). A control mutation takes the behaviour path outside the safe region.

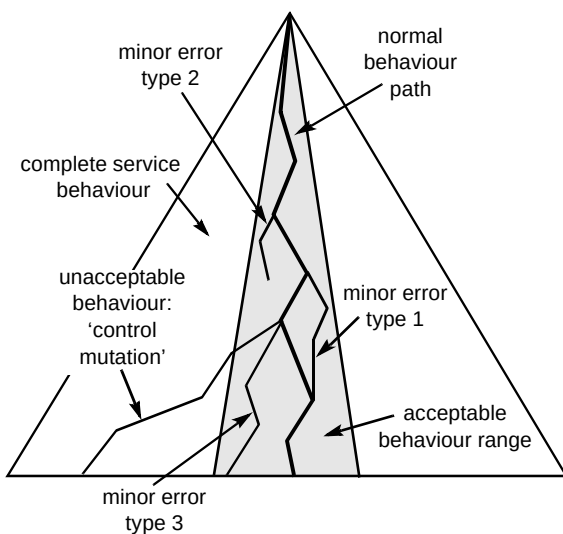


Fig 4 Path of correct behaviour and deviations from it.

The stages to develop the framework, for the simplified case of one single service, are presented next and illustrated in Fig 5.

9.2 Stage 1 — modelling and understanding of service behaviour

A formal 'service description' that defines the behaviour and operation of the service is needed, e.g. SDL can be used to describe the overall behaviour, whereas an MSC [15] can be used to visualise selected traces and identify message interchanging between communicating entities.

9.3 Stage 2 — finding path of normal behaviour

Given a formal description of a particular service — such as an SDL description of a VPN (virtual private network) service — one path of desired behaviour is focused on and the transactions involved are analysed, e.g. signalling, state transitions, etc, from the MSC description.

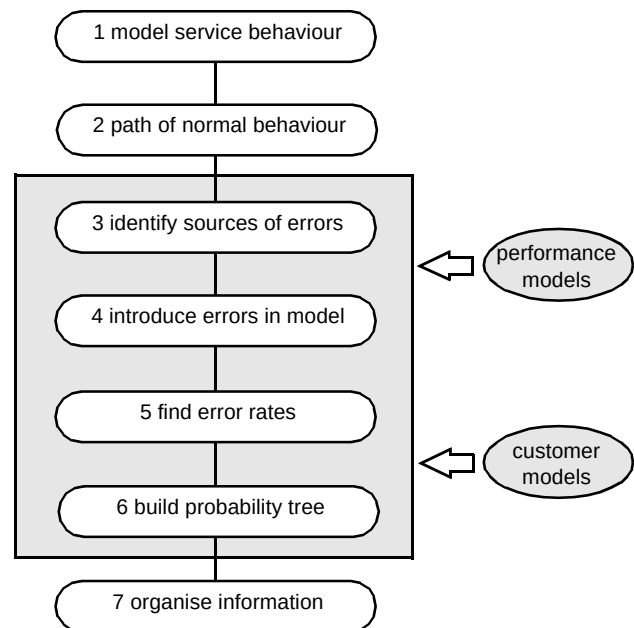


Fig 5 Stages of the proposed methodology

9.4 Stage 3 — identification of possible sources of error

After describing the behaviour, it is necessary to identify possible error events, e.g. perturbations such as invalid messages that may trigger a violation of integrity, and study how the introduction of these errors affects the behaviour of the service.

Two inputs are needed in order to identify errors, namely performance models and population models.

Performance models

As previously stated (section 4), performance characteristics can have a great impact on the behaviour of services. Two types of errors can be differentiated, namely:

- static errors, due to bad definition of services, poor logic or incorrect programming, inherent to the services, which can be minimised with adequate specifications, precise definitions and carefully developed programming,
- dynamic errors, which occur due to performance limitations from delays, signalling overloads, etc, even when a system is correctly developed.

Dynamic errors are more difficult to detect and correct because they are caused by numerous event combinations. In order to identify sources of dynamic errors performance models need to be built and coupled to the overall model. One of the major limitations of SDL in this context is that it does not include performance evaluation (some research is being done in that direction [20]). There is a need to study

how to couple the performance models to the SDL description of the service behaviour.

Customer models

The importance of customer models, e.g. usage patterns and demand characteristics, was discussed in section 4. These customer models are needed as another input to the overall model.

Sensitivity analysis must be carried out on performance and population models to find the limits for which they are valid.

By using formal services descriptions, performance characteristics of the systems and population models to build a generic model, the output at the end of this stage will be the identification of a set of errors that can occur within a given environment.

9.5 Stage 4 — introduction of errors in the model and identification of error states

The MSCs that describe a normal path of behaviour need to be modified to introduce errors. Analysis of this information will permit the identification of error states that can be gradually incorporated to the model in order to build paths of erroneous behaviour.

9.6 Stage 5 — finding probabilities for the different error states

After a list of possible error events has been identified, they must be matched to rates, i.e. the probability of errors occurring per time unit. Performance characteristics and population models need to be taken into account to obtain the probabilities of each particular type of error. Initially, error rates must be assigned based on external information, e.g. previous experiences, and there will be an associated degree of uncertainty. This uncertainty will always exist, but it will be reduced as the methodology progresses and provides more information about errors and their probabilities.

9.7 Stage 6 — construction of overall probability tree

The worst problems are likely to arise when a chain of errors occur, hence the need to identify possible sequences of error states that will progressively take the system away from the normal behaviour. The correlation between different error states, i.e. the probability that one error state leads to another, forms a large and complex decision tree. A simple example is demonstrated by the tree structure illustrated in Fig 6. Each node represents a state and the

interconnecting branches represent transitions between states and their probability. This will show the overall probability of moving from the initial state to each of the possible final states.

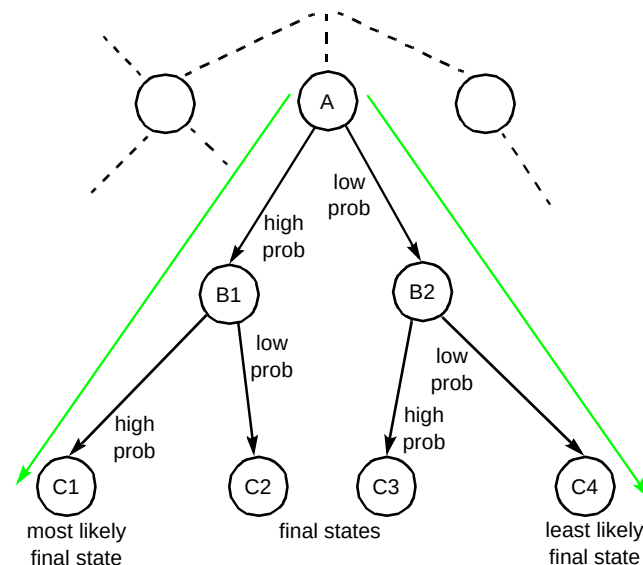


Fig 6 Example of state transitions and probabilities.

In complex systems, where the number of possible paths and final states is extremely large, it would be unfeasible to find the probabilities to reach each of the states. Therefore, the work must concentrate on two cases, namely:

- the most likely states (such as state C1), which describe the normal operation of the system,
- the most dangerous states, i.e. those that fall out of the safe region, even if their occurrence is very unlikely, because they need to be quickly identified in order to prevent catastrophic failures.

Regardless of the extreme improbability of entering a dangerous control sequence, identification of such sequences is extremely important. This is because a system operating for a significant length of time will explore a considerable fraction of all the possible sequences. Hence it will eventually enter this dangerous region with catastrophic consequences. In summary, unlikely events are inevitable, provided enough time elapses, and if such unlikely events lead to a threat to integrity, the threat will manifest itself. The key point is to rapidly identify them and take the appropriate actions before the system falls over.

9.8 Stage 7 — organisation of information

The last stage of the methodology is to extract useful information from the data obtained in the previous stages, so that it can be easily accessed and utilised as a pre-emptive framework. There is the need for adequate

documentation of all the acquired information, possibly in the form of a new kind of expert system where it can be extracted in an intelligent way.

10. Conclusions

The ever-increasing complexity of telecommunications networks and services progressively heightens the probability of service outages due to loss of integrity. The paper has outlined a methodology that can be developed to determine the cause, extent and outcome of integrity violations, thus assisting the design of robust systems and the management of 'live' network services. At present, the methodology is being tested for the simplified case of one service and one network. SDL models of the service are being developed. Future work will include more complex scenarios, i.e. multiservice, multi-network, thus allowing the examination of issues such as service interaction and interconnect.

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